Google Cloud Platform Deployment Guide

Overview

This guide provides step-by-step instructions for deploying the ROSHN Parametric Masterplan Modelling Platform to Google Cloud Platform (GCP) using Cloud Run and Cloud SQL.

Architecture

```
┌─────────────────────────────────────────────────────────────────┐
│                     GOOGLE CLOUD PLATFORM                        │
│                   Region: me-central1 (Dammam)                   │
├─────────────────────────────────────────────────────────────────┤
│                                                                  │
│   ┌──────────────┐     ┌──────────────┐     ┌──────────────┐   │
│   │   Internet   │────▶│  Cloud Run   │────▶│  Cloud SQL   │   │
│   │              │     │ (Container)  │     │ (PostgreSQL) │   │
│   └──────────────┘     └──────────────┘     └──────────────┘   │
│                               │                    ▲             │
│                               │                    │             │
│                               ▼                    │             │
│                        ┌──────────────┐           │            │
│                        │ VPC Connector │───────────┘            │
│                        └──────────────┘                         │
│                               │                                  │
│                               ▼                                  │
│                        ┌──────────────┐                         │
│                        │Secret Manager │                         │
│                        └──────────────┘                         │
│                                                                  │
│   ┌──────────────────────────────────────────────────────┐     │
│   │                  Artifact Registry                    │     │
│   │                   (Docker Images)                     │     │
│   └──────────────────────────────────────────────────────┘     │
│                                                                  │
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```

Prerequisites

Required Tools

Tool	Purpose
Google Cloud SDK (gcloud)	GCP command-line interface
Docker	Container build and push
Git	Source code management

Required Permissions

- Project Owner or Editor role
- Billing account linked to project

GCP Services Used

Service	Purpose
Cloud Run	Serverless container hosting
Cloud SQL	Managed PostgreSQL database
Artifact Registry	Docker image storage
Secret Manager	Secure credential storage
VPC Access Connector	Private network connection
Cloud Build	Container building

Deployment Steps

Step 1: Install Google Cloud SDK

Download and install from: https://cloud.google.com/sdk/docs/install

Verify installation:

```
gcloud version
```

Step 2: Authenticate with GCP

```
gcloud auth login
```

Step 3: Clone Repository

```
git clone https://github.com/tajalagawani/roshn.git
cd roshn
```

Step 4: Run Deployment Script

Run the deployment script with your target environment:

```
chmod +x scripts/deploy-gcp.sh

# Deploy to specific environment
./scripts/deploy-gcp.sh dev      # Development environment
```

```
./scripts/deploy-gcp.sh staging  # Staging environment
./scripts/deploy-gcp.sh prod     # Production environment (default)
```

Environment Configurations

Environment	Cloud SQL Tier	Cloud Run Memory	Instances	Description
dev	db-f1-micro	512Mi	0-2	Scale to zero, minimal cost
staging	db-custom-1-3840	1Gi	0-5	Testing environment
prod	db-custom-2-4096	2Gi	1-10	Always-on production

Configuration

Default Configuration Values

Update these values in `deploy-gcp.sh` before running:

Variable	Default Value	Description
PROJECT_ID	roshn-platform	GCP project ID
REGION	me-central1	Deployment region (Dammam)
APP_NAME	roshn-platform	Application name
DB_INSTANCE_NAME	roshn-db-instance	Cloud SQL instance name
DB_NAME	roshn_platform	Database name
DB_USER	roshn_admin	Database username

Cloud Run Configuration

Setting	Default Value	Description
MEMORY	2Gi	Container memory allocation
CPU	2	Number of vCPUs
MIN_INSTANCES	1	Minimum running instances
MAX_INSTANCES	10	Maximum instances for scaling

Cloud SQL Configuration

Setting	Default Value	Description
DB_TIER	db-custom-2-4096	2 vCPU, 4GB RAM

PostgreSQL Version	14	Database version
Storage Type	SSD	High-performance storage
Storage Size	50GB	Initial storage (auto-increase enabled)

What the Script Does

1. Project Setup

- Creates or selects GCP project
- Enables required APIs:
    - Cloud Build API
    - Cloud Run API
    - Cloud SQL Admin API
    - Secret Manager API
    - Artifact Registry API
    - Compute Engine API
    - VPC Access API

2. Artifact Registry

- Creates Docker repository for container images
- Configures Docker authentication

3. Cloud SQL Instance

- Creates PostgreSQL 14 instance
- Configures automated backups (3:00 AM daily)
- Sets maintenance window (Sunday 4:00 AM)
- Enables deletion protection
- Creates database and user

4. VPC Connector

- Creates serverless VPC access connector
- Enables private communication between Cloud Run and Cloud SQL

5. Secret Manager

- Stores database connection URL securely
- Stores NextAuth secret for authentication

6. Docker Image

- Creates production Dockerfile
- Builds optimized container image
- Pushes to Artifact Registry

7. Cloud Run Deployment

- Deploys container to Cloud Run
- Configures environment variables
- Sets up Cloud SQL connection
- Enables public access

8. Database Migrations

- Runs Prisma migrations via Cloud Run jobs
- Seeds initial data

Post-Deployment

Access Your Application

After deployment, the script displays your application URL:

```
Application URL: https://roshn-platform-xxxxx-me.a.run.app
```

View Logs

```
gcloud run services logs read roshn-platform --region=me-central1
```

Update Application

After code changes:

```
# Build and push new image
docker build -t me-central1-docker.pkg.dev/roshn-platform/roshn-platform/roshn-platform:latest .
docker push me-central1-docker.pkg.dev/roshn-platform/roshn-platform/roshn-platform:latest

# Deploy new version
gcloud run deploy roshn-platform \
    --image=me-central1-docker.pkg.dev/roshn-platform/roshn-platform/roshn-platform:latest \
    --region=me-central1
```

Connect to Database

```
gcloud sql connect roshn-db-instance --user=roshn_admin --database=roshn_platform
```

Custom Domain Setup

Step 1: Navigate to Cloud Run Domains

Go to: https://console.cloud.google.com/run/domains

Step 2: Add Domain Mapping

1. Click "Add Mapping"
2. Select the roshn-platform service
3. Enter your custom domain
4. Follow DNS configuration instructions

Step 3: Update DNS Records

Add the provided DNS records to your domain registrar:

- CNAME or A record as instructed by GCP

Scaling Configuration

Automatic Scaling

Cloud Run automatically scales based on traffic:

Setting	Configuration
Scale to zero	Disabled (MIN_INSTANCES=1)
Maximum instances	10
Concurrency	80 requests per instance

Modify Scaling

```
gcloud run services update roshn-platform \
    --region=me-central1 \
    --min-instances=2 \
    --max-instances=20
```

Security Features

Built-in Security

Feature	Status
HTTPS	Automatic (managed certificates)
Secret Management	Secret Manager integration
Private Database	VPC connector (no public IP)
IAM	Service account permissions

Environment Variables

Sensitive values are stored in Secret Manager:

- DATABASE_URL
- NEXTAUTH_SECRET

Monitoring

Cloud Console

- **Cloud Run**: https://console.cloud.google.com/run
- **Cloud SQL**: https://console.cloud.google.com/sql
- **Logs**: https://console.cloud.google.com/logs

CLI Commands

```
# Service status
gcloud run services describe roshn-platform --region=me-central1

# Recent logs
gcloud run services logs read roshn-platform --region=me-central1 --limit=50

# Database status
gcloud sql instances describe roshn-db-instance
```

Troubleshooting

Common Issues

Container fails to start

- Check logs: `gcloud run services logs read roshn-platform --region=me-central1`
- Verify environment variables are set correctly

Database connection errors

- Verify VPC connector is working
- Check Cloud SQL instance is running
- Verify connection string in Secret Manager

Build failures

- Check Dockerfile syntax
- Ensure all dependencies are in package.json
- Verify Node.js version compatibility

Get Help

```
# Describe service for detailed status
gcloud run services describe roshn-platform --region=me-central1 --format=yaml

# List recent revisions
gcloud run revisions list --service=roshn-platform --region=me-central1
```

Cost Management

Key Cost Factors

Service	Billing Model
Cloud Run	Per request + CPU/memory time

Cloud SQL	Instance size + storage + network
Artifact Registry	Storage used
VPC Connector	Per hour while active

Cost Optimization Tips

1. **Scale to zero** - Set MIN_INSTANCES=0 for dev/test
2. **Right-size database** - Start with smaller tier, scale up as needed
3. **Clean old images** - Delete unused container images
4. **Use committed use discounts** - For production workloads

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Useful Commands Reference

Command	Description
`gcloud run services list`	List all Cloud Run services
`gcloud run services logs read SERVICE`	View service logs
`gcloud run services update SERVICE`	Update service configuration
`gcloud sql instances list`	List Cloud SQL instances
`gcloud sql connect INSTANCE`	Connect to database
`gcloud secrets list`	List all secrets
`gcloud artifacts docker images list`	List container images